



The impact of judicial protection of intellectual property on digital innovation: Evidence from China

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ABSTRACT

Using the patent data of Chinese A-share listed companies from 2009 to 2020 and the pilot of China's intellectual property (IP) courts in late 2014, we apply a triple difference (DDD) method to estimate the effect of judicial protection of IP on corporate digital innovation. Our findings suggest that enhanced judicial protection of IP has a positive impact on corporate digital innovation, and that it contributes more to underlying digital technology innovation. Moreover, small firms are more likely to be affected by judicial protection of IP in their digital innovation activities than large firms.

1. Introduction

Digital innovation refers to the integration of digital technology and physical components of products in the innovation process, including the creation of new products, improvements in production processes, and changes in business models (Nambisan et al., 2017; Fichman et al., 2014; Yoo et al., 2012). However, unlike traditional innovations, digital innovation products are primarily composed of computer codes, which are easily replicable, and susceptible to reverse engineering and functional reconstruction (Boudreau et al., 2022; Liu et al., 2020; Miric et al., 2019), exacerbating the risk of digital innovation spillovers and making it difficult for firms to benefit from digital innovation (Teece, 2018). More seriously, China's legal system is facing institutional issues (Yang et al., 2023), including inconsistent judicial standards, complex litigation procedures, and low compensation in IP cases, all of which further discourage firms from innovating digitally.

According to the existing literature, strong judicial protection can effectively defer imitation (Kong et al., 2023; Neves et al., 2021; Meng et al., 2021), increase firms' marginal profit from innovation, and provide dynamic incentives for investment in innovation (Lin, 2023; Sampat and Williams, 2019). However, the measurement of judicial protection of IP in previous studies mainly focused on two levels: legal construction and the intensity of law enforcement leading to potential endogenous problems such as measurement errors and underrepresentation of samples (Li et al., 2021; Feng et al., 2017). In fact, strengthening the judicial protection of IP depends not only on the construction of the legal system and the intensity of law enforcement, but also on improving enforcement efficiency (Ang et al., 2014). Therefore, in this context, how strengthening the judicial protection of IP will affect enterprises' digital innovation needs to be further explored.

To improve the efficiency of IP enforcement, many developed countries (such as the UK and Japan) have established IP courts. In this context, China, the largest innovative developing country, established IP courts in Beijing, Shanghai and Guangzhou in 2014 to

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establish a platform to maintain a common dialogue with the international community and to strengthen domestic IP protection. In the context of China's unique innovation environment, the IP Courts have carried out a substantial number of reforms in a short period, boldly exploring numerous new mechanisms for IP litigation, addressing institutional issues that hinder judicial efficiency (Zhang et al., 2022; Genin et al., 2022; Li et al., 2021), and optimizing the judicial environment for digital innovation. For instance, the Beijing IP Court issued the "Reference of Proof in Civil Cases of Infringement of Trade Secrets"; the Shanghai IP Court actively explored the "Internet+" trial model, holding pretrial conferences for technical investigations and trials via remote video; and the Guangzhou IP Court formulated the "Guidelines on the Application of Punitive Damages" to increase the penalties for serious infringements such as repeated infringement and malicious infringement.

Currently, the literature has explored various factors that influence digital innovation, including digital strategy (Wang et al., 2022), digital resources (Hanelt et al., 2021; Firk et al., 2022), and digital innovation capabilities (Svahn et al., 2017). Some scholars focus on industry status (Liu et al., 2021), digital innovation culture (Lokuge et al., 2019), and user feedback (Trinugroho et al., 2022; Abrell et al., 2016). However, Meng et al. (2021) and Boudreau et al. (2022) analyzed the protection of IP strategies of firms with different digital innovation achievements, but the research on the impact of judicial protection of IP on digital innovation in enterprises is still limited. Therefore, we aim to fill this gap by using a DDD model to evaluate the impact of establishing IP courts on digital innovation among China's A-share listed companies from 2009 to 2020.

Our study contributes to the relevant literature in several ways. Firstly, we use the DDD model to address the endogenous problem between the establishment of IP Courts and firms' digital innovation, laying the foundation for an accurate assessment of the impact of the establishment of IP courts on firms' digital innovation. Secondly, we enrich the study on the economic consequences of IP Courts and expand the exploration of the factors that influence digital innovation. From the perspective of judicial efficiency of IP, we find that the establishment of IP Courts can significantly promote corporate digital innovation. Finally, we conduct an in-depth study on the relationship between IP Courts and corporate digital innovation, dividing the latter into underlying digital technology innovation and digital technology application innovation. We further divide the sample into large firms and small firm for deeper investigation. Our findings provide insights into the judicial reform of IP and incentives for firms to innovate digitally.

2. Research design

2.1. Sample selection

We select data of Chinese A-share listed companies from 2009 to 2020, obtain the text data of corporate patent abstract from the Wingo financial text data platform, and obtain financial data from the CSMAR database. We exclude companies in the ST, finance and insurance industries from the sample, and finally obtain 28,959 firm-year observations. Descriptive statistics for all variables are shown in Appendix A.

2.2. Regression model

To test the impact of judicial protection of IP on firms' digital innovation, we construct the following difference-in-difference (DID) model:

$$DI_{i,t+1} = \alpha_0 + \alpha_1 Treat_i \times Post_t + \alpha_2 Treat_i + \alpha_3 Post_t + \beta Controls_{i,t} + \delta_{Industry} + \varphi_{Year} + \varepsilon_{i,t} \quad (1)$$

where i and t represent firm and year, respectively. Dependent variable DI is corporate digital innovation. Considering that the impact of variables is time-consuming in practice, we advance DI by one period. Following the approach of Firk et al. (2022) and Wang et al. (2022), we use text analysis to search for company patent abstracts containing 76 keywords developed by Wu et al. (2021) from the two dimensions of underlying digital technology and digital technology applications (see Appendix B), and aggregate patents containing the above keywords by year. Since the data is "right skewed", we add one and take the natural logarithm to obtain the more accurate index of DI describing the company's digital innovation.

The core explanatory variable $Treat \times Post$ is the interaction term between the treatment variable $Treat$ and the establishment of IP courts $Post$. The treatment group includes companies with offices in Beijing, Shanghai, and Guangdong (excluding Shenzhen), where $Treat$ takes the value of 1 and 0 otherwise. $Post$ is a dummy variable representing before and after the establishment of IP courts, where $Post$ takes the value of 1 for the post-2014 samples and 0 otherwise. To avoid the interference of corporate financial and governance characteristics on our results, we control for relevant variables defined in Appendix C. In addition, we control for time and industry fixed effects as random disturbances. To address firm-level heteroskedasticity, the regressions below cluster firm codes and use robust standard errors.

As IP Courts are located in first-tier cities with developed economies and concentrated high-tech industries, such as Beijing, Shanghai, and Guangzhou, local R&D in digital technology is more intensive, and enterprises are more sensitive to judicial protection of IP. In this context, using the DID model may lead to inconsistent estimation results. Therefore, we introduce a new variable in model (2), namely whether it is a high-tech industry ($Tech$), and construct the following DDD model.

$$DI_{i,t+1} = \alpha_0 + \alpha_1 DDD + \delta X + \beta Controls_{i,t} + \delta_{Industry} + \varphi_{Year} + \varepsilon_{i,t} \quad (2)$$

where DDD is the interaction term of $Treat$, $Post$, and $Tech$ variables. X is the interaction term of any two of the three variables $Treat$, $Post$, and $Tech$, and other variables are consistent with model (1). According to the *Guidelines for Industry Classification of Listed*

Companies (2012 Revision), we define industry classification codes C25-C29, C31-C32, C34-C41, I63-I65 and M73 as high-tech industries, which mainly involve manufacturing, information transmission, software, and information technology services. When the enterprise belongs to high-tech industry, the value of Tech is 1, and 0 otherwise.

3. Empirical results

3.1. Baseline results

Table 1 reports the baseline regression results of the impact of judicial protection of IP on firms' digital innovation. Columns (1)-(2) and (3)-(4) present the estimates obtained by the DID and DDD methods, respectively. Columns (1) and (3) do not include time and industry fixed effects. The results confirm that regardless of the method used, the coefficients of the core interaction terms are significantly positive at the 1% level, indicating that the establishment of IP Courts promotes digital innovation in firms. Specifically, the coefficient of the interaction term in column (2) is 0.089, implying that the establishment of IP Courts increases the number of digital innovations in firms by 9.31% ($e^{0.089}-1$); the coefficient of the interaction term in column (4) is 0.163, implying that the establishment of IP Courts increases digital innovations in high-tech industries by 17.70% ($e^{0.163}-1$).

3.2. Parallel trend test

To test the parallel trend hypothesis, we compare the time trends of digital innovation among firms in the treatment and control groups. Fig. 1 shows that prior to the establishment of IP Courts (i.e., before 2015), the trends in digital innovation are similar in both groups. However, after the establishment of IP Courts (i.e., after 2014), there is a substantial change in the trends of digital innovation in both groups, suggesting that the sample data basically meets the premise conditions for using the multiple difference method.

Table 1
Baseline regression results.

Variables	(1) DI _{it,t+1}	(2) DI _{it,t+1}	(3) DI _{it,t+1}	(4) DI _{it,t+1}
DDD			0.173*** (3.552)	0.163*** (3.399)
Treat _{it} × Post _{it}	0.089*** (4.441)	0.089*** (4.501)	0.036** (2.231)	0.035** (2.138)
Treat _{it} × Tech _{it}			0.018 (0.527)	−0.079** (−2.205)
Post _{it} × Tech _{it}			0.083*** (5.230)	0.068*** (4.258)
Treat _{it}	−0.010 (−0.636)	−0.018 (−1.161)	−0.014 (−1.104)	0.013 (1.049)
Post _{it}	0.056*** (6.513)	0.047*** (3.676)	0.019** (2.510)	0.022* (1.951)
Tech _{it}			0.110*** (5.584)	0.097*** (4.449)
Size _{it}	0.079*** (6.243)	0.078*** (6.172)	0.076*** (6.122)	0.075*** (6.063)
Age _{it}	−0.004*** (−4.120)	−0.001 (−1.072)	−0.004*** (−3.607)	−0.001 (−1.234)
Lev _{it}	−0.063* (−1.706)	0.031 (0.861)	−0.010 (−0.259)	0.044 (1.218)
Roa _{it}	0.354*** (2.664)	0.528*** (3.927)	0.335*** (2.594)	0.525*** (3.984)
Cash _{it}	0.255*** (4.371)	0.164*** (2.846)	0.223*** (4.070)	0.160*** (2.844)
Fixed _{it}	−0.249*** (−6.039)	−0.235*** (−4.641)	−0.263*** (−6.284)	−0.194*** (−3.987)
Inst _{it}	−0.136*** (−4.597)	−0.059** (−2.188)	−0.098*** (−3.614)	−0.054** (−2.047)
Top10 _{it}	0.058** (2.236)	0.038 (1.521)	0.060** (2.378)	0.039 (1.577)
Constant	−1.574*** (−5.958)	−1.748*** (−5.204)	−1.585*** (−6.021)	−1.670*** (−5.016)
Year	No	Yes	No	Yes
Industry	No	Yes	No	Yes
N	24822	24822	24822	24822
Adj-R2	0.056	0.128	0.093	0.142

Note: ***, **, and * represent 1%, 5%, and 10% significance levels, respectively. Variables are clustered by firm code, and robustness is used to adjust for standard errors.

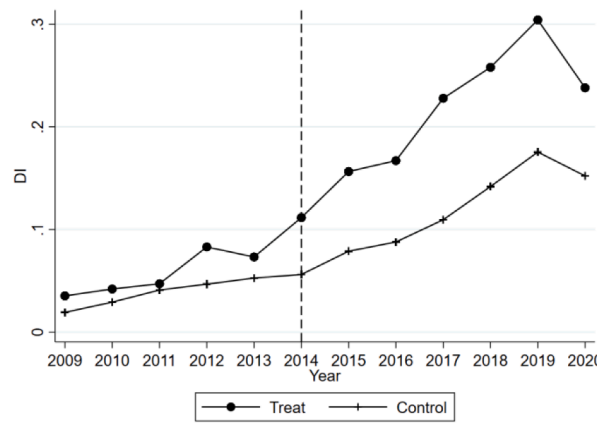


Fig. 1. The parallel trend of digital innovation.

3.3. Robustness test

To control and eliminate selectivity bias, we use the propensity score matching method by using the control variables in model (1) as matching variables and conducting a 1:1 without replacement nearest neighbor matching. The estimation results are presented in columns (1)-(2) of Appendix D. To avoid endogeneity problems caused by omitted variables, we further control for firm fixed effects in model (1) and (2), and the estimation results are presented in columns (3)-(4) of Appendix D. It can be observed that the coefficients of the core interaction terms are positive and significant, further demonstrating the robustness of our findings.

In addition, since 2017, the Supreme People's Court of China has approved the establishment of 26 local IP Tribunals. To avoid the interference from the establishment of these tribunals on the causal identification of our study, we utilize the following staggered DID and DDD models.

$$DI_{i,t+1} = \alpha_0 + \alpha_1 Post1_{i,t} + \phi Controls_{i,t} + \delta_{Industry} + \varphi_{Year} + \varepsilon_{i,t} \quad (3)$$

$$DI_{i,t+1} = \alpha_0 + \alpha_1 DDD1 + \alpha_2 Post1_{i,t} + \alpha_3 Tech + \phi Controls_{i,t} + \delta_{Industry} + \varphi_{Year} + \varepsilon_{i,t} \quad (4)$$

where dummy variable $Post1_{i,t}$ is based on whether the city to which enterprise i belongs is within the jurisdiction of the IP Court (Tribunal) in year t . $DDD1$ is the intersection of $Post1$ and $Tech$. The estimation results of models (3) and (4) are shown in columns (5)-(6) of Appendix D. It can be observed that the core explanatory variables remain significantly positive at the 1% level, and our conclusion remains unchanged.

3.4. Heterogeneity analysis

To further analyze the impact of IP Courts on firms' digital innovation, we differentiate digital innovation into two categories: (1) underlying digital technology innovation and (2) digital technology application innovation. To measure underlying digital technological innovation, we classify patents containing 42 keywords related to artificial intelligence, blockchain, cloud computing, and big data in the patent abstracts as underlying digital technological innovation $DI1$. Patents abstracts containing any of the other 34 keywords in Appendix B are classified as digital technology application innovation $DI2$. We use $DI1$ and $DI2$ as explanatory variables in model (2), and the estimation results are shown in columns (1)-(2) in Appendix E. It can be found that the coefficient of the interaction term DDD is significantly positive at the 1% level in column (1), but not significant in column (2), indicating that the establishment of IP Courts mainly promotes underlying digital technology innovation.

Compared to large firms, small firms may face greater external financial constraints due to lack of collateral and are less likely to use expensive legal procedures to protect their IP (Narayan et al., 2023; Chen et al., 2023). Therefore, the establishment of IP Courts is expected to have a greater impact on digital innovation in small firms. We divide the sample into two subsamples of large and small firms based on firms' average size and estimate model (2) for each subsample. The estimation results are shown in columns (3)-(4) of Appendix E. It can be found that the regression coefficient of the interaction term DDD is significantly positive in small firms, but not significant in large firms, which is consistent with our expectation.

4. Conclusion

This study focuses on Chinese A-share listed companies from 2009 to 2020 and uses text analysis methods and corporate patent data to measure corporate digital innovation. By constructing a DDD model, we explore the impact of IP Courts on firms' digital innovation and its heterogeneity. Our findings indicate that judicial protection of IP has a significant positive impact on firms' digital

innovation, which is supported by a series of robustness tests. In addition, we find that judicial protection of IP has a stronger promotion effect on underlying digital technology innovation. At the firm level, small firms are more susceptible to the impact of judicial protection of IP on digital innovation than large firms. This study emphasizes that the establishment of IP Courts as a judicial system reform can stimulate digital innovation in enterprises and provide insights for government to implement innovation-driven development strategies.

CRedit authorship contribution statement

Panpan Zheng: Methodology, Data curation, Software, Writing – original draft. **Zhen Li:** Software, Validation, Writing – review & editing. **Ziyin Zhuang:** Conceptualization, Supervision, Funding acquisition.

Data availability

Data will be made available on request.

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Appendix

Appendix A

Descriptive statistics of main variables.

Variables	Obs	Mean	SD	Min	Median	Max
DI	28959	0.112	0.483	0.000	0.000	6.368
Treat	28959	0.234	0.424	0.000	0.000	1.000
Policy	28959	0.610	0.488	0.000	1.000	1.000
Tech	28959	0.407	0.491	0.000	0.000	1.000
Size	28959	22.146	1.287	19.917	21.954	26.164
Age	28959	9.653	7.355	0.000	8.000	26.000
Lev	28959	0.420	0.207	0.050	0.413	0.879
Roa	28959	0.041	0.056	−0.217	0.039	0.195
Cash	28959	0.193	0.141	0.017	0.153	0.689
Fixed	28959	0.215	0.162	0.002	0.182	0.700
Inst	28959	0.448	0.247	0.003	0.470	0.908
Top10	28959	0.354	0.228	0.008	0.343	0.860

Appendix B

Digital innovation keywords.

Variables	Dimension	Keywords
Underlying digital technology	Artificial Intelligence Technology	Artificial intelligence, business intelligence, image understanding, investment decision aid system, intelligent data analysis, intelligent robotics, machine learning, deep learning, semantic search, biometrics, face recognition, voice recognition, identity verification, autonomous driving, natural language processing
	Big Data Technology	Big data, data mining, text mining, data visualization, heterogeneous data, credit, augmented reality, mixed reality, virtual reality
	Cloud Computing Technology	Cloud computing, streaming computing, graph computing, in-memory computing, multi-party secure computing, brain-like computing, green computing, cognitive computing, converged architecture, billion level concurrency, EB level storage, Internet of Things, information physical systems
	Blockchain Technology	Blockchain, digital currency, distributed computing, differential privacy technology, smart financial contracts
Digital Technology Applications		Mobile Internet, industrial Internet, mobile Internet, Internet healthcare, e-commerce, mobile payment, third-party payment, NFC payment, smart energy, B2B, B2C, C2B, C2C, O2O, Netlink, smart wear, smart agriculture, smart transportation, smart healthcare, smart customer service, smart home, smart investment advisor, smart cultural tourism, smart environmental protection, smart grid, smart marketing, digital

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Variables	Dimension	Keywords
		marketing, unmanned retail, Internet finance, digital finance, Fintech, financial technology, quantitative finance, open banking

Appendix C

Variable definitions.

Variables	Definition
DI	Digital innovation
Treat	Treatment variable
Policy	Establishment of Intellectual Property Court
Tech	High-tech industry
Size	Firm scale
Age	Firm Age
Lev	Gearing ratio
Roa	Profitability
Cash	Cash Flow
Fixed	Fixed assets
Inst	institutional investors
Top10	Equity concentration

Appendix D

Robustness tests.

Variables	(1) PSM DI _{i,t+1}	(2) DI _{i,t+1}	(3) FE DI _{i,t+1}	(4) DI _{i,t+1}	(5) IP Court (Tribunal) DI _{i,t+1}	(6) DI _{i,t+1}
DDD		0.097 (1.446)		0.098* (1.747)		
Treat _{i,t} × Post _{i,t}	0.084*** (3.017)	0.057** (2.487)	0.063*** (2.758)	0.031* (1.655)		
DDD1						0.124*** (4.050)
Post1 _{i,t}					0.058*** (3.402)	0.013 (0.851)
Treat _{i,t} × Tech _{i,t}		−0.050 (−1.202)		−0.015 (−0.362)		
Post _{i,t} × Tech _{i,t}		0.119** (2.520)		0.094*** (5.087)		
Treat _{i,t}	−0.034** (−1.979)	−0.013 (−0.797)	−0.057 (−1.329)	−0.054 (−1.334)		
Post _{i,t}	0.001 (0.045)	−0.040 (−1.568)	0.389 (0.655)	0.348 (0.622)		
Tech _{i,t}		0.061* (1.835)		−0.079*** (−4.802)		0.107*** (5.398)
Size _{i,t}	0.089*** (4.093)	0.081*** (3.843)	0.039*** (3.458)	0.031*** (2.884)	0.079*** (6.186)	0.076*** (6.092)
Age _{i,t}	−0.002 (−0.938)	−0.002 (−0.999)	−0.073 (−0.615)	−0.071 (−0.637)	−0.001 (−1.082)	−0.001 (−1.294)
Lev _{i,t}	−0.068 (−0.912)	−0.051 (−0.680)	0.072** (2.320)	0.072** (2.334)	0.032 (0.883)	0.049 (1.330)
Roa _{i,t}	0.445** (2.032)	0.466** (2.147)	0.167** (2.469)	0.159** (2.355)	0.519*** (3.858)	0.529*** (3.989)
Cash _{i,t}	0.136 (1.417)	0.152 (1.618)	−0.059* (−1.711)	−0.029 (−0.840)	0.164*** (2.840)	0.154*** (2.715)
Fixed _{i,t}	−0.166* (−1.801)	−0.118 (−1.240)	−0.000 (−0.002)	−0.001 (−0.026)	−0.238*** (−4.786)	−0.197*** (−4.167)

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Variables	(1) PSM DI _{i,t+1}	(2) DI _{i,t+1}	(3) FE DI _{i,t+1}	(4) DI _{i,t+1}	(5) IP Court (Tribunal) DI _{i,t+1}	(6) DI _{i,t+1}
Inst _{i,t}	−0.068 (−1.458)	−0.064 (−1.399)	−0.110*** (−2.953)	−0.103*** (−2.848)	−0.060** (−2.209)	−0.055** (−2.084)
Top10 _{i,t}	0.032 (0.599)	0.041 (0.781)	−0.016 (−0.945)	−0.008 (−0.504)	0.038 (1.492)	0.037 (1.492)
Constant	−1.859*** (−4.144)	−1.707*** (−3.966)	−0.069 (−0.064)	0.117 (0.115)	−1.767*** (−5.174)	−1.702*** (−5.025)
Firm	No	No	Yes	Yes	No	No
Year	Yes	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	No	No	Yes	Yes
N	7314	7314	24822	24822	24822	24822
Adj-R2	0.160	0.174	0.035	0.043	0.127	0.141

Note: ***, **, and * represent 1%, 5%, and 10% significance levels, respectively. Variables are clustered by firm code, and robustness is used to adjust for standard errors.

Appendix E

Heterogeneity analysis.

Variables	(1) Underlying digital technology DI1 _{i,t+1}	(2) Digital technology applications DI2 _{i,t+1}	(3) Large firms DI _{i,t+1}	(4) Small firms DI _{i,t+1}
DDD	0.112*** (3.344)	0.018 (1.102)	0.070 (0.976)	0.299*** (4.661)
Treat _{i,t} × Post _{i,t}	0.022** (2.236)	0.003 (0.447)	0.044 (1.506)	0.002 (0.138)
Treat _{i,t} × Tech _{i,t}	−0.048** (−2.151)	−0.022* (−1.730)	−0.124 (−1.366)	−0.051* (−1.718)
Post _{i,t} × Tech _{i,t}	0.049*** (4.364)	0.009 (1.252)	0.010 (0.275)	0.065*** (3.597)
Tech _{i,t}	0.049*** (3.662)	0.039*** (4.246)	0.227*** (3.810)	0.052*** (3.434)
Treat _{i,t}	0.008 (0.921)	0.003 (0.704)	0.024 (1.000)	−0.004 (−0.464)
Post _{i,t}	0.017* (1.893)	0.001 (0.264)	0.034 (1.438)	0.027** (2.141)
Size _{i,t}	0.044*** (5.574)	0.021*** (4.677)	0.138*** (4.905)	0.035*** (3.784)
Age _{i,t}	−0.001 (−1.317)	−0.000 (−0.618)	−0.003 (−1.338)	−0.001 (−1.039)
Lev _{i,t}	0.025 (1.092)	0.003 (0.319)	0.092 (1.293)	0.049 (1.594)
Roa _{i,t}	0.290*** (3.389)	0.156*** (3.194)	1.167*** (3.999)	0.251** (2.427)
Cash _{i,t}	0.091** (2.558)	0.033* (1.872)	0.312** (2.118)	0.085** (2.092)
Fixed _{i,t}	−0.118*** (−3.710)	−0.052*** (−3.028)	−0.289*** (−3.355)	−0.142*** (−4.313)
Inst _{i,t}	−0.027 (−1.631)	−0.020*** (−2.593)	−0.143** (−2.193)	−0.030 (−1.347)
Top10 _{i,t}	0.023 (1.507)	0.013 (1.472)	0.084** (2.049)	−0.009 (−0.436)
Constant	−1.094*** (−6.131)	−0.493*** (−4.906)	−3.032*** (−4.969)	−0.698*** (−3.664)
Year	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes
N	24822	24822	10777	14045
Adj-R2	0.107	0.041	0.166	0.155

Note: ***, **, and * represent 1%, 5%, and 10% significance levels, respectively. Variables are clustered by firm code, and robustness is used to adjust for standard errors.

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